

CSA0315 – DATA STRUCTURE
CO3 – AT2 (TREE TRAVERSAL VISUALIZATION)
VISUALIZE TREE TRAVERSAL

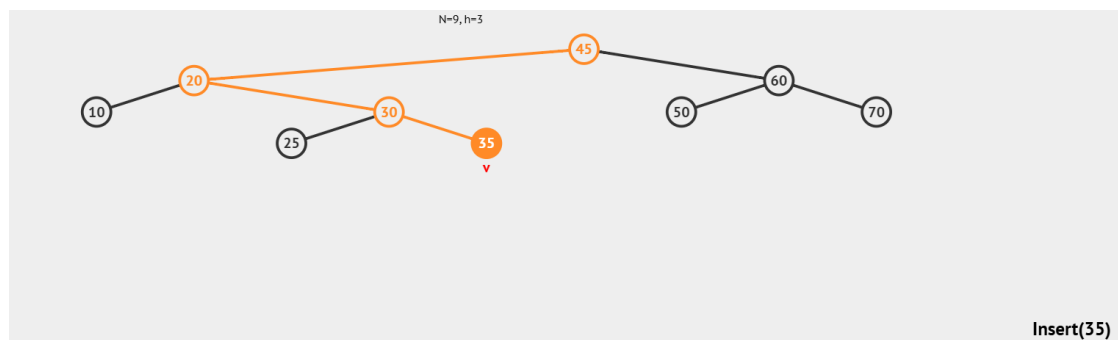
Faculty:

Dr. T. Rajesh Kumar

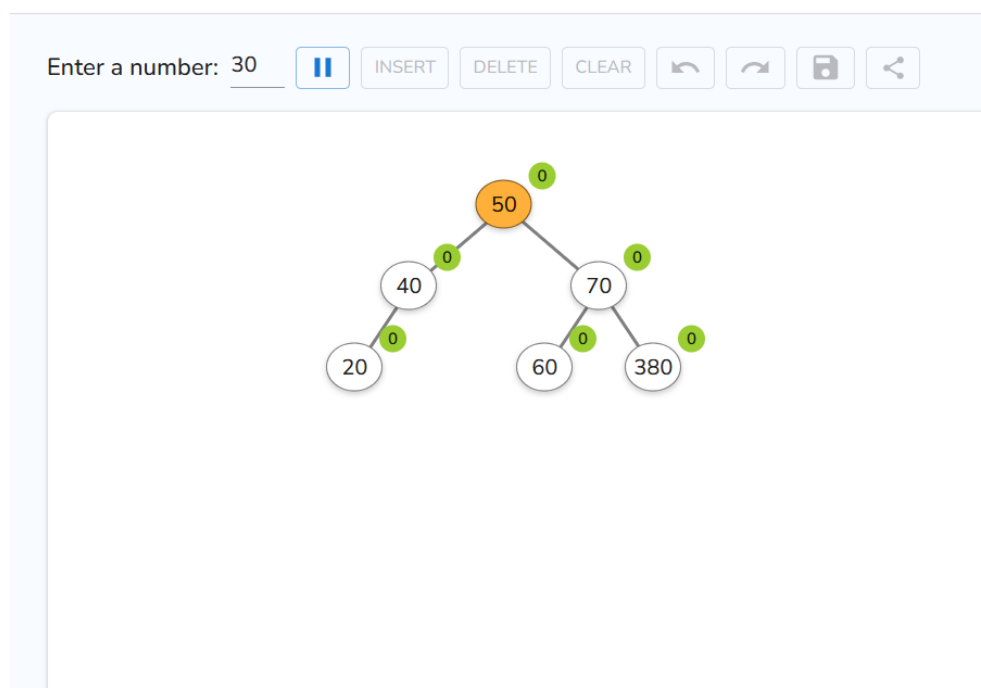
By:

Harshith G (192525394)

1) Construct a Binary Search Tree (BST) using the dataset 45,20,60,10,30,50,70,25,35



2) Construct an AVL Tree using the elements 50,30,70,20,40,60,80 perform the required operations and document the results. Delete node 30



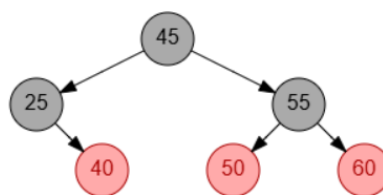
3) Construct a Red-Black Tree using {45, 35, 55, 25, 40, 50, 60}, delete the node 35, and explain how violations are fixed using rotations and recoloring

Insert

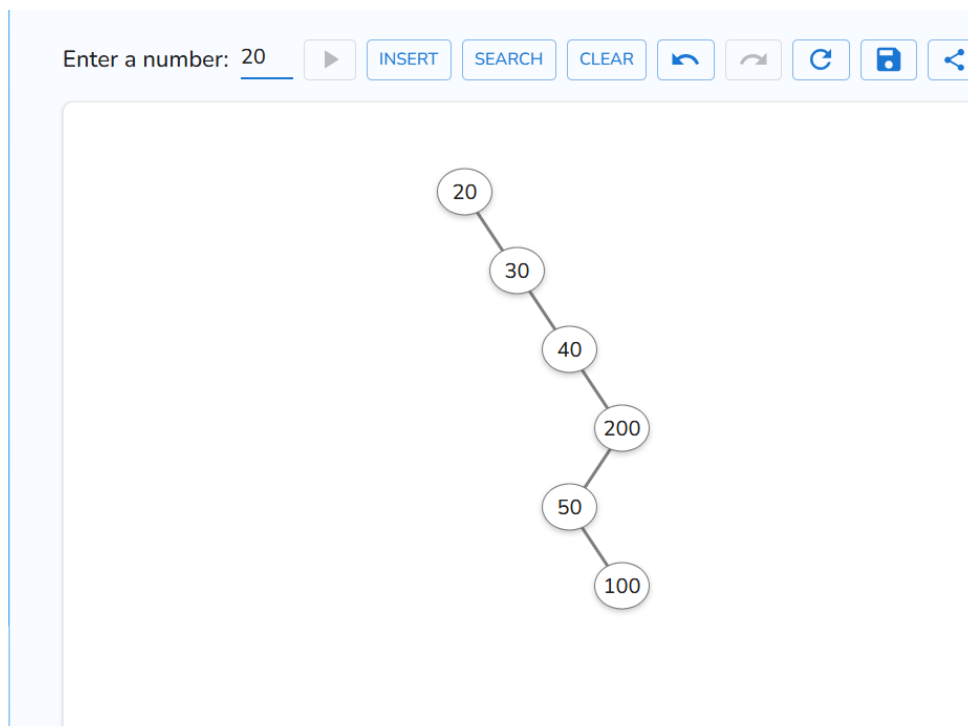
Insert

Delete

Delete



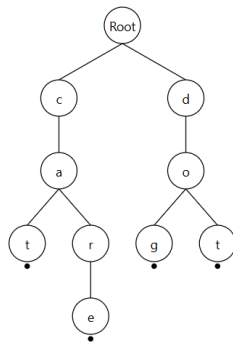
4) Develop a Splay Tree with elements {100, 50, 200, 40, 30, 20}, perform a search operation for 20, and demonstrate the zig-zig rotations that bring it to the root



- 5) Create a 2-3-4 Tree for the dataset {50, 40, 60, 30, 70, 20, 80}, perform insertions step-by-step, and show how 4-nodes are split during the process.



- 6) Construct a Trie using the words {cat, car, care, dog, dot}, perform insertion for all words, search for the prefix “ca”, and delete the word “car” while maintaining the Trie structure.



Insert

Delete

Search

Random Initialize

Words in Trie

care

cat

dog

dot